How to Dilute Penicillin G to PCN Allergy Testing

(Using a 5,000,000 unit vial)

Step 1: Reconstitute vial per instructions on outer packaging

- Three options are recommended by the manufacturer depending on desired concentration.
- Choose the one that requires the least amount of volume (less manipulations)
    - Add 3.2 mL of diluent (either Sterile Water for injection or sterile Normal Saline)
    - Yield = **1,000,000 units/mL** (or 1MU/mL)

Step 2: Dilute reconstituted product to desired concentration of **10,000 units/mL**

- Take 0.1 mL of the 1MU/mL and QS to **10mL** with 9.9mL of normal saline to get the desired concentration of **10,000 units/mL**.

Explanation and rationale for dilution in PEN Allergy Test:

- A. Determine the **total volume of the final dilution you want**.
- B. Using $C_1V_1 = C_2V_2$, determine how much volume of your current dilution you will need.
- C. You should only need about 0.1mL for each of the two intradermal injections (about 0.2mL total per test). Therefore, 10mL is enough for about 50 patients, and is a practical quantity for measuring/diluting.

- Example: If we choose to make **10mL** of the desired **10,000 units/mL** penicillin G
    - Dilute Pen G vial to 1MU/mL (as above)
    - Using $C_1V_1 = C_2V_2$
        - (1MU/mL) (V_1) = (10,000 units/mL) (10mL)
        - V_1 = 0.1mL

- Therefore, you would take 0.1 mL of the 1MU/mL and QS to **10mL** with 9.9mL of normal saline to get the desired concentration of **10,000 units/mL**.
- Draw up 0.2 ml into 1 ml TB syringes.
- Expiration: Six months